

VANTARA PETROCHEM - UNIT 3

PRESSURE VESSEL INSPECTION REPORT

Report Number:	INS-2023-V310-001
Equipment Tag:	V-310
Equipment Type:	Pressure Vessel - Overhead Accumulator Drum
Inspection Date:	2023-08-15
Inspector:	Arvind Menon (EMP-005)
Inspection Type:	External Visual + UT Thickness Survey

THICKNESS SURVEY RESULTS

Location	Nominal (mm)	Measured (mm)	% Loss	Status
Shell Course 1 - 0°	12.0	11.8	1.7%	ACCEPTABLE
Shell Course 1 - 90°	12.0	11.7	2.5%	ACCEPTABLE
Shell Course 1 - 180°	12.0	11.9	0.8%	ACCEPTABLE
Shell Course 1 - 270°	12.0	11.6	3.3%	ACCEPTABLE
Shell Course 2 - 0°	12.0	11.5	4.2%	ACCEPTABLE
Shell Course 2 - 90°	12.0	10.2	15.0%	REQUIRES ACTION
Shell Course 2 - 180°	12.0	11.4	5.0%	ACCEPTABLE
Shell Course 2 - 270°	12.0	10.8	10.0%	MONITOR
Head - Top	14.0	13.8	1.4%	ACCEPTABLE
Head - Bottom	14.0	13.6	2.9%	ACCEPTABLE

FINDINGS AND OBSERVATIONS

CRITICAL FINDING - CUI Detected:

Location: Shell Course 2, 90° position (North side)
Insulation was removed for inspection at suspected CUI location. Significant external corrosion observed under insulation. Wall thickness reduced to 10.2mm from nominal 12.0mm, representing 15% wall loss. This exceeds the 10% threshold per OISD-STD-119 Section 5.1.4 requiring engineering assessment within 30 days.

Root Cause Assessment: Moisture ingress through damaged insulation jacket at pipe support location. Galvanic corrosion accelerated by dissimilar metal contact with carbon steel support clip.

Secondary Finding:

Location: Shell Course 2, 270° position

Wall loss at 10% - at threshold level. Recommend monitoring at 6-month interval.

RECOMMENDATIONS

IMMEDIATE (within 30 days):

1. Engineering assessment required per OISD-STD-119 for 15% wall loss finding
2. Repair or replace insulation jacket at affected location
3. Install insulation standoff at pipe support to eliminate galvanic couple
4. Consider protective coating application to affected area

MONITORING:

1. Re-inspect 270° location at 6-month interval
2. Add V-310 to CUI susceptibility monitoring program

Note: Work order must be raised within 7 days per compliance requirements to initiate corrective action for critical finding.

CERTIFICATION

This inspection was performed in accordance with API 510 and company procedures. The equipment is fit for continued service provided the recommendations above are implemented within the specified timeframes.

Inspector: Arvind Menon, API 510 Certified

Date: 2023-08-15

Next Inspection Due: 2024-08-15 (or upon completion of repairs, whichever is earlier)